MATCH WORDS WITH DEEDS: AN AI-EMPOWERED PROXY FRAMEWORK FOR ORGANIZATIONAL MICROFOUNDATIONS

Background

A central topic in organization research is to unveil organizational microfoundations—defined as the cognitions, beliefs, and traits of individual members, as well as the collection and integration thereof, which can steer firms' strategic decisions, operations, and performance (Barney & Felin, 2013; Felin et al.,, 2015). However, as these microfoundational features essentially entail latent cognitive and psychological properties, they are generally hard to observe and quantify. Prior organization studies have primarily resorted to two streams of proxies to measure microfoundations. The first stream, which tends to be more traditional, strives to extract quantitative cues from firms' financial reports and disclosures—R&D expenditures, acquisition premiums, lobbying spending, overheads, and so on—as ***behavioral proxies*** for the focal firms' latent microfoundations (e.g., Coles et al., 2006; Hoskisson et al., 1993). The second stream draws on the contents, rhetorics, and sentiments of firms' public communications. Scholars extract the linguistic features of these texts to develop ***narrative proxies*** that reflect the firms' latent microfoundations (Loughran & McDonald, 2011; Tetlock et al., 2008). To construct such proxies, scholars either compile static dictionaries of keywords or adopt natural language processing (NLP) techniques to feature the themes of corporate documents.

These two streams both entail a key underlying assumption: firms with particular microfoundational features would be systematically more likely to make certain decisions, engage in certain operational activities, and frame the narratives of their communications in certain ways. However, in practice, this assumption can be subject to inherent challenges that threaten the validity and power of both streams of proxies.

A key challenge commonly facing behavioral proxies is the problem of attribution: a firm's strategic decisions and operational activities can be determined by many factors, especially the inherent specificities in the focal firm and its environments. As such, many behavioral cues can only be partially accounted for by the firm's microfoundational features, thus creating major confounds in these behavioral proxies.

Such confounds also affect exist in narrative proxies. In firms' narratives, many microfoundational cues are intertwined with other contents. These contents, as the contexts of narrative cues, do not directly reflect the firms' microfoundations. However, it is often hard for scholars to accurately tease apart those contextual contents from valid microfoundational cues. Moreover, firms can often strategically orchestrate and manipulate the contents and rhetorics of public communications due to social desirability and impression management, thus rendering those narratives at variance with the focal firms' true beliefs and cognitions. As such, the validity of narrative proxies can often be challenged by the problem of authenticity.

In sum, both streams of microfoundational proxies face potential measurement errors, which could largely dampen the empirical and theoretical rigorousness of microfoundations research. Addressing this gap urges organizational scholars to explore nuanced measurements to more effectively identify the underlying microfoundations beyond the proxy power afforded by mere behavioral or narrative cues.

Mission of this study

In this study, we strive to develop a nuanced AI-empowered framework to integrate the behavioral and the narrative cues for the same latent microfoundational feature to develop a synthesized proxy thereof. To accomplish this task, we integrate three core AI techniques, i.e., unsupervised machine learning (ML) (*topic modeling*), LLM agents, and supervised ML (*gradient boosting and regularized regression*), to develop a scalable workflow. The nuanced proxy framework can effectively extract the narrative cues in corporate texts and documents that are closely related to a given microfoundational feature, map such narrative cues with established behavioral proxies to extract the predicted common variation thereof, and validate the power of such predicted common variation as the synthesized proxy for the microfoundational feature.

Basic workflow

Step 1: Characterizing the Corpus

As the majority of microfoundations research emphasizes both within- and between-unit differences, the first step needs to establish a generalizable narrative framework that can describe the features of all firms' corporate texts with the universal scale. Following the mainstream practice, we use unsupervised topic discovery (Blei et al., 2003; Grootendorst, 2022) to establish the thematic structure emerging from the distributional properties of corporate linguistics. We adopt the word embedding-based approach to cluster transformer representations over the semantic space. By doing so, a firm's corporate texts is described by the dense domain vocabulary and the meaning that depends on context, rather than on the presence of particular words (Li et al., 2021). In this step, the unsupervised topic discovery is applied to the corpus in its entirety and remains microfoundation-agnostic, yielding the topic structure that is the input to the second stage.

Step 2: Topic decontextualization

This step converts the corpus-wide topic structure into a subject-specific measurement directly pertaining to the focal microfoundational feature in a study. The task here is one of selection and interpretation—establishing which topics constitute evidence about the focal feature rather than about the contextual specificities facing a firm. We address it through decontextualization. Each topic is evaluated for the extent to which it is subject-specific or context-dependent. A subject-specific topic's contents bear directly on the focal feature, such that the topic itself constitutes evidence about the firm's microfoundational characteristics; a context topic describes the setting in which the firm operates—its unique resource pool, its industry, its competitive environment, its market conditions. Only subject-specific topics enter the narrative proxy. The judgment is made by multiple LLM agents operating over the full topic structure established in Step 1. We use peer votes as the judgement threshold: agents first classify topic descriptors, then review common source passages within their own histories, and an unweighted two-of-three majority determines selection. Because divergence is expected to concentrate on topics that describe an activity the firm's microfoundation would govern while remaining a description of the activity itself, the agents' concurrence functions as an accuracy gauge internal to the workflow, generated by the process that produces the labels rather than imposed on it afterward.

Step 3: Synthesize the narrative and behavioral proxies

After determining the list of subject-specific topics, we compute for each retained topic two critical qualities: *density,* the share of the firm's communication devoted to that topic, which measures emphasis rather than content; and *sentiment*, the direction and tone of the firm's discussion of that topic, which measures orientation rather than extent. The conjunction of these two qualities constitutes the feature-relevant cues. We compile a vector of topic density and sentiment in the corporate texts of each sample firm each year, which serves as the ground for its narrative proxy for the focal microfoundational feature.

This narrative proxy (in terms of the density-sentiment vector) is then used to predict an established behavioral proxy for the focal microfoundational feature. We adopt *XGBoost*, a supervised ML algorithm entailing the regularized form of gradient boosting that incorporates second-order (Hessian) information in both the leaf-weight solution and the split-gain criterion, along with Ridge regression, to perform such prediction. The fitted value of the behavior for each observation serves as the synthesized proxy. This proxy, which entails the behavioral outcome as reconstructed from the narrative evidence, effectively captures the covariation between the narrative proxy and the behavioral proxy. The residual of the fitted model, in this regard, records for each observation how far its behavior departs from its narratives (i.e., decoupling).

Step 4: Validate the synthesized proxy

Step 4 strives to gauge the synthesized proxy in terms of both convergent validity and incremental validity. Convergent validity is measured with the extent to which the synthesized proxy is systematically correlated with, and can predict, both the narrative proxy and the behavioral proxy. Evidence of this kind

establishes consistency with an alternative measurement rather than with the latent ground truth. Incremental validity indicates whether the synthesized proxy can afford extra explanatory power beyond the narrative proxy and the behavioral proxy together. Evidence of this kind licenses a falsifiable proposition rather than a claim of superiority. Its value lies as much in its failure modes as in its success, because each outcome carries a specific implication. If the synthesized proxy performs no better than the narrative proxy, the behavioral mapping has added nothing and existing text-based measures are adequate. If it performs no better than the behavioral proxy, the narrative evidence contributes nothing the established determinants of behavior do not already carry, and measuring the focal feature from language is not supported. If it outperforms both, the framework's premise is corroborated on evidence external to its own construction. This is what makes the claim falsifiable rather than rhetorical: it specifies in advance the conditions under which it would be wrong. Evidence for such incremental validity can generally be derived from replicating prior studies on the same microfoundational feature.

Preliminary validation

We have conducted a demo study to validate the feasibility of the above workflow. In this demo study, we focused on a representative microfoundational feature that has been widely examined in prior studies, i.e., CEOs' risk preference (e.g., Coles et al., 2006; Li & Tang, 2010; Wiseman & Bromiley, 1996). We use CEOs' linguistics in conference calls as the ground to develop the narrative proxy. Using *BERtopic*, we extract 354 topics across all corpus inputs. The votes of three LLM agent readers (powered by a frontier LLM; model, reasoning effort, prompts, and voting rule documented in the technical appendix) eventually identified 321 topics as related to risk preference. Retention is deliberately lenient at this final stage: descriptor-only screening retains 66–87 topics per run, and selection rises to 88.4–90.7% only after the readers review source passages. We then adopt *XGBoost* and Ridge regression to fit the density-sentiment vector of the 321 topics with a well-established behavioral proxy for CEO risk preference, i.e., R&D intensity. The fitted value is used as the synthesized proxy to replicate the results of Caliskan and Doukas (2015), a widely cited study on the implications of CEO risk preference with all data publicly accessible.

Please see the ***technical appendix*** and ***GitHub repo*** (https://github.com/jianghan6666/worddeed) for details of the demo study. As can be seen, the synthesized proxy of CEO risk preference afforded satisfactory levels of convergent validity and incremental validity, providing preliminary evidence for the validity and feasibility of our AI-empowered framework to proxy organizational microfoundations.

Key risks, future work, and implications

There are several important risks that we plan to address in our development next.

First, our framework essentially strives to identify the common variation between the behavioral proxy and the narrative proxy of the same microfoundational features as the synthesized proxy. As denoted by the congeneric measurement model (Graham, 2006), a key assumption needs to stand for this synthesized proxy to afford higher power than either proxy alone: This common variation between the behavioral and narrative proxies needs to be primarily caused by the latent microfoundations as their shared driver, so that the two proxies' error variance are independent. The decontextualization in Step 2 is critical to defend this assumption, such that it removes contextual texts that may carry inherent specificities that could steer firms' behaviors. In the development next, we plan to specifically gauge the rigorousness of the synthesized proxy by adding variance decomposition as a standing robustness check, recovering the common-factor share of the synthesized proxy's variances as the indicator of potential proxy failure.

Another risk is that the LLM judgments on which topic selection rests may prove unstable across model versions and prompting configurations, so that the measurement pipeline embeds a moving target. Our experiments (see appendix) corroborated such result drifting across different models. We plan to address

this risk through version pinning, reporting and analyzing a complete validation record covering model versions, prompts, reasoning effort, and source context, and using such report as a robustness gauge.

Along with the above directions, we will also further corroborate the feasibility and validity of our framework with more microfoundational features at various levels and with different sources of behavioral and narrative cues (in this regard, we have already conducted another trial with political donation records and developed a synthesized proxy for corporate political ideology, which also demonstrated satisfactory convergent and incremental validity).

Implications. If the framework works, for microfoundations research, it offers a design template that confronts a central empirical challenge: the inherent measurement error, thus helping enhance the consistency between theory and empirics in this field. In a broader sense, the design of this framework also provides a scalable and inspectable toolkit for organizational research, which allows AI judgments to function as explicit, versioned, and falsifiable measurement inputs rather than opaque intermediaries.

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FIGURE 1

Constructing and testing a synthesized proxy

Researchers define the focal microfoundation and the unit of analysis.

